

AI AssemblyTwin — Detailed Business Proposal

Accenture Innovation Challenge 2026 · Round 2
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Executive Summary

AI AssemblyTwin is a production-grade digital twin of a 45-station vehicle assembly line. It combines real-time SimPy simulation, four trained ML models, and a full-stack web application to deliver predictive quality and maintenance intelligence — before defects reach the customer. The system directly addresses all four constraints in the Round 2 brief: uneven sensor coverage, multi-causal defect origins, read-only OPC-UA architecture, and multi-stakeholder dashboards. Payback period: 2.5 months. 3-year net ROI: ■6.1 Crore per plant.

Problem Statement Alignment

Constraint	Our Solution
Uneven sensor coverage	Gaussian Process imputes missing readings in real-time with explicit uncertainty
Multi-causal defect origins	Random Forest with lag features traces upstream causality: <code>/api/causal/chair</code>
No direct PLC modifications	Read-only OPC-UA model; all interventions require human supervisor approval
Multi-stakeholder views	4-tab Analytics: Floor Supervisor · Plant Manager · Leadership · ESG

Technical Architecture

Data Flow:

SimPy Assembly Simulation (45 stations) → FastAPI Backend (4 ML models, 10 API endpoints, WebSocket) → Next.js 14 Frontend (6 pages, real-time data binding, Framer Motion animations).

ML Model Details:

Model	Algorithm	Key Innovation	Output
Anomaly Detector	Isolation Forest	Unsupervised — no labelled data needed	$\text{anomaly_score} \in [-1,1]$
Bottleneck Predictor	LSTM (2-layer)	Converted to NumPy — <100MB RAM footprint	$\text{bottleneck_station} \in [0,1]$
Defect Predictor	Random Forest 200t	Lag features capture upstream causality	$\text{defect_risk} + \text{feature importance}$
Sensor Imputer	Gaussian Process	Explicit uncertainty for legacy stations	$\text{value} + \sigma$

UI/UX Design Decisions

Before/After Toggle: Toggle Digital Twin OFF → shows cost of inaction (■48L rework, 4.2hr downtime). Toggle ON → shows what the twin prevented. Instantly quantifies ROI for any audience.

WHY? Explainability: Anomalous stations show a WHY? button. Clicking it shows SHAP-style feature importance in plain English. Addresses the black-box AI concern in 10 seconds.

Predictive Maintenance Calendar: Monthly view with colour-coded dates. Proactive vs reactive cost comparison banner (8.3× multiplier). Turns abstract ML into an operational schedule.

Live ROI Ticker: Ticks up in █ at █800/minute baseline. Jumps by █3.5L on every approved intervention. Makes savings tangible in real-time during a live demo.

Multi-Site Network: 3-plant overview proves enterprise scalability from day one. Not a lab prototype.

Stakeholder Value

Stakeholder	Key Value
Floor Supervisor	Real-time anomaly map + WHY? explainability + one-click intervention approval
Plant Manager	7-day throughput trends + bottleneck frequency + maintenance calendar
Leadership / CXO	Interactive ROI calculator + 3-plant network + ESG impact metrics
CSR / Sustainability	CO█, steel, energy saved per session + UN SDG 9/12/13 alignment

ROI Model

Item	Value	Basis
Deployment cost	█45 L	Infrastructure + 4-week integration
Defect cost avoided	█3.5 L/event	Industry avg rework cost per escaped defect
Throughput gain	+4.2%	Bottleneck removal via proactive scheduling
Monthly savings	█18 L	Blended: defect prevention + throughput + maintenance
Payback period	~2.5 months	█45L ÷ █18L
3-year ROI	█6.1 Cr net	36 × █18L – █45L
Emergency repair multiplier	8.3×	Reactive vs proactive industry benchmark

ESG & Sustainability Case

Per prevented scrap vehicle: 333 kg CO█ avoided · 180 kg steel saved · 12L paint solvent avoided · 0.4 kWh energy saved.

UN SDG Alignment: SDG 9 (Industry & Innovation) · SDG 12 (Responsible Consumption) · SDG 13 (Climate Action).

Accenture has committed to Net Zero by 2025. Every prevented scrap vehicle is a measurable, auditable CO█ reduction that directly supports client ESG reporting.

Implementation Timeline

Phase	Duration	Deliverable
Data & Simulation	Week 1–2	SimPy model, synthetic dataset, OPC-UA schema

ML Training	Week 2–3	All 4 models trained, validated, serialised
Backend API	Week 3–4	FastAPI + WebSocket, all 10 endpoints
Frontend UI	Week 4–6	All 6 pages, real-time integration
Integration & Testing	Week 6–7	E2E demo, load testing, edge case hardening
Deployment	Week 8	Docker compose, documentation, handover

Team

- Jayanth — AI & System Architecture
- Abhinav — Industrial Integration & Telemetry
- Sagar — Frontend UX & Data Analytics

GitHub: github.com/jay-git00/DigitalTwin.ai